AUTOMATED MANUFACTURING SYSTEMS

OVERVIEW

Applying leadership and 21st century skills, participants collaborate to design and prototype an automated manufacturing system to solve a problem. Teams analyze the prompt, build a sensor-based mechatronic solution, program the automation logic, and document their design. Finally, teams create instructions for judges to operate the system.

ELIGIBILITY

Two (2) teams of two to three (2-3) members per state may participate.

TIME LIMITS

A. Competition consists of three (3) phases.
   1. Phase 1: Setup – Teams are allowed twenty (20) minutes to setup their equipment.
   2. Phase 2: Planning – Teams are allowed twenty-five (25) minutes for project analysis, brainstorming, and planning.
   3. Phase 3: Building – Teams are provided two and a half (2.5) hours to develop and program their solution.

B. Teams participate in demonstration and a five (5) minute interview.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.

B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.

C. Teams arrive at the time and location stated in the conference program. Once teams are checked in they will be allowed twenty (20) minutes to set up their equipment.

D. At the conclusion of the setup phase, teams will receive the problem statement and will be allowed twenty-five (25) minutes for planning. During the planning phase, teams will only be allowed to discuss with each other, and make drawings or notes using paper and pencil. No other equipment may be used. Note: Student planning notes are not assessed on the rubric.

E. Each team is given a maximum of two and a half (2.5) hours to:
   1. Integrate mechanical components and sensors to create a functional prototype.
   2. Program the system logic required to execute the task.
   3. Debug the system through iterative testing, to ensure reliability and repeatability.
   4. Complete the “Engineer’s Guide” by documenting the system logic and operating instructions.

F. Demonstration and Interview:
   1. Safe State: When the build phase ends, teams must save their code, place their system in a “Safe/Home” position, and leave the code visible on the laptop screen.
   2. Functional Demonstration: With judges present, the team initiates the system. The system must execute two (2) or more full cycles (as defined in the problem statement) autonomously.

      Note: Teams may interact with the laptop and/or Human-to-Machine interface (e.g., button) to trigger the initial start cycle, or any specific action in the on-site problem statement.
   3. Interview: Directly after the demonstration, judges will conduct a five (5) minute interview, focusing on the automation logic, mechanical design choices, and troubleshooting steps.

G. Judges independently assess the entries, including each team’s interview responses.

H. The top ten (10) finalists are announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

A. No step-by-step build instructions or pre-written programs are allowed. Teams are permitted to have physical or digital reference materials (e.g., syntax guides, wiring diagrams, datasheets) to aid in building or programming on-site.

B. Participants provide their own laptop and programming software. Note: Internet connection is not provided by TSA, locally installed "off-line" software is recommended. Laptop must be labeled with team identification number, advisor name, and advisor telephone number on the top.

C. Teams are required to provide their own box(es)/container(s), which must meet the following requirements:

   1. Tool box/container must be labeled with team identification number, advisor name, and advisor telephone number on the top.
   2. Tool box/container must not exceed 20" x 20" x 40". If multiple tool boxes/containers are used, the combined dimensions cannot exceed 20" x 20" x 40".
   3. Laptops, charger, and paper/pencil can be carried separately and are not part of the size constraint.
   4. Contents: Tool box/container must include all items needed to fabricate a solution. To ensure teams can solve the problem, each kit must contain at least:
      a. Sensors to detect an object's presence and an object's properties (e.g., color, touch, distance, optical, etc.).
      b. Motors or servos sufficient to power a drive train and/or stationary mechanisms.
      c. Devices to produce distinct light outputs (e.g., LEDs or stack lights) and devices to produce audible sound (e.g., speaker).
      d. Drive train components (e.g., gears, wheels, sprockets, chain, axles, etc.).
      e. Distinct colored objects: balls, blocks, cylinders to be moved, sorted, and manipulated. Have twelve (12) objects of each type. Each type of object should have three different colors (e.g., four (4) red blocks, four (4) green blocks, four (4) blue blocks). 3D printed objects are OK to use.
      f. Structural components (e.g., beams, plates, fasteners, etc.) to build a rigid device.

D. No cutting devices or power tools may be used; materials must retain the original form in which they were brought to the competition.

E. No adhesives such as glue or tape is allowed.

F. Each team must have a power strip with surge protector and a 6" or more extension cord.

G. The following definitions are an integral part of the event regulations:

   1. Continuous Cycle: The system is programmed to reset automatically after a process is complete, readying itself for the next step without human intervention.
   2. Autonomous Operation: The system must accomplish the task efficiently without human interaction. The logic should handle decision-making (e.g., sorting based on sensor data) rather than relying on humans.
   3. Lean Design: The device reflects the efficient use of materials to solve the problem. The design should be robust but not overbuilt, minimizing the physical footprint and material waste.

H. Programs must be written entirely on-site.

   1. Use or modification of any pre-written program files, code blocks, or sub-routines created prior to the competition is prohibited and will result in disqualification.
   2. Teams may view their reference materials (PDFs/syntax guides) but must manually type or construct the logic during the event time limits.

I. Two (2) example problem statements for this event are provided below. Note: These example problem statements provide the overall challenge but not the full requirements of the problem.

Example 1:
A manufacturing company needs an automated system to catch defective parts before they reach packaging.

Build a prototype for a "Smart Quality Inspection Station." Parts must travel down a conveyor belt into an inspection zone. A sensor must identify defective items (such as the wrong color or size). Defective parts must be automatically pushed off the line into a reject bin, while good parts continue forward. To save power, the conveyor motor should only run when a part is detected at the start of the line.

Example 2:
Amusement Park Boarding Station Simulation Problem: An amusement park engineering firm requires a prototype for an automated roller coaster boarding station. To ensure passenger safety, the mechanical boarding gates must remain closed while the coaster car is out on the track. Once the car returns and comes to a complete stop, the gates must open to allow passengers to board. To prevent premature coaster launch while passengers are boarding, a boarding operator must manually press a button to close the gates before the coaster can automatically leave for its next run.

EVALUATION

A. Written work
B. Model function
C. Programming structure and efficiency
D. Interview

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Use of Generative Artificial Intelligence (GenAI) is NOT permitted for this event.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Manufacturing Automation Engineer
- Mechatronics Technician
- PLC Programmer
- Robotics Engineer
- Systems Integrator
- Control Systems Engineer
- Electro-Mechanical Technician

ENGINEER'S GUIDE

TEAM ID

System Logic: List the specific Inputs (Sensors) and Outputs (Motors/Lights) your solution uses and define their function in your solution.

EXAMPLE

Device Type	Port/ID	Function / Logic Assignment
INPUT (Optical Sensor)	Port 1	IF optical sensor sees a RED block, THEN stop conveyor and signal Port 2.
OUTPUT (Motor)	Port 2	WHEN signaled by Port 1, rotate arm to push block into scrap bin.

Operating Instructions: Provide clear, step-by-step instructions for the evaluator to operate your system.

Participant/Team ID# _____

AUTOMATED MANUFACTURING SYSTEMS

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Laptop is present
☐ Tool box/container is present
☐ ENTRY NOT EVALUATED

ENGINEER'S GUIDE (20 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
System Logic (X1)	System logic section is incomplete; inputs or outputs are missing; function descriptions are vague (e.g., "Motor Moves") or do not clearly define the trigger/action relationship; written plan does not match the physical model.	System logic section includes most necessary inputs and outputs with accurate port IDs; function descriptions are generally clear, though some "If/Then" logic may be simplified; plan accurately reflects the physical model's operation.	System logic section is a comprehensive and precise guide of all Inputs/Outputs and port IDs; logic assignments are written in clear "If/Then" conditional statements that perfectly describe the system's behavior; plan shows foresight into efficient programming.	
Operating Instructions (X1)	Instructions are vague, disorganized, or incomplete; key steps for powering on or resetting the system are missing; evaluator cannot reliably operate the system using only the written guide.	Instructions are functional and generally clear; evaluator can successfully start and reset the system, though some steps may be slightly ambiguous or require interpretation.	Instructions are professional, concise, and "foolproof"; step-by-step protocols for both start-up and reset are clearly defined, allowing the evaluator to operate the system flawlessly without clarification.	
			ENGINEER'S GUIDE SUBTOTAL (20 points)	

SOLUTION TO PROBLEM (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Realistic Simulation (X1)	Design resembles a loose collection of parts rather than a cohesive automated system; lacks operational realism (e.g., sensors or mechanisms are placed haphazardly with no clear logic to system flow or user interaction).	Design logically simulates an automated system or process; components are arranged in a functional sequence that mimics real-world operation, though spatial organization or physical constraints could be improved.	Design serves as a highly realistic simulation of a fully integrated automated system; layout is optimized for efficient workflow or user interaction, and component placement reflects professional engineering practices.	
Mechanical Reliability (X1)	Structure is flimsy or unstable; vibration from motors causes sensors to misalign or parts to shift during operation; mechanism jams frequently or requires human adjustment.	Structure is generally stable; drive train and mechanisms function reliably for most cycles, though minor adjustments may be needed between runs.	Structure is rigid and well constructed; drive trains and manipulators operate with high precision and repeatability; system creates no excessive vibration and handles mechanical stress without failure.	
Lean Design (X1)	Design is "overbuilt" with unnecessary beams or plates that clutter the workspace; wire routing is messy, creating potential snag hazards; inefficient use of materials is obvious.	Materials are used effectively to create a functional system; design is not excessively bulky; wire management is adequate, keeping cables generally clear of moving parts.	Design exemplifies "Lean" principles; uses the minimum amount of material necessary to achieve structural rigidity; wire routing is clean, bundled, and professional, ensuring safety and aesthetics.	
Function (X2)	System fails to perform the core task(s); multiple required attributes (e.g., rejection of bad parts) are missing or non-functional; cycle does not reset.	System performs the core task(s) with reasonable success; most attributes function as intended, though the timing or speed may be inefficient.	System flawlessly executes all core task(s) and required attributes; cycle runs smoothly from start to finish and resets automatically without error.	
Innovation and Creativity (X1)	Solution relies on the simplest possible mechanisms; design shows little attempt to use mechanical advantage, gearing, or complex motion.	Solution uses standard mechanical concepts; design is effective but standard for the challenge.	Solution demonstrates advanced mechanical insight; team utilizes complex mechanisms to achieve smoother or faster operation than standard designs.	
			SOLUTION TO PROBLEM SUBTOTAL (60 points)	

PROGRAMMING STRUCTURE (20 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Control Logic (X1)	Program relies heavily on "Hard-Coded" timing (e.g., "Wait 5 seconds") rather than sensor feedback; code is linear and repetitive, failing to use loops or variables to simplify tasks.	Program uses sensor feedback to trigger most actions; logic follows the standard "If/Then" structure; code is readable, though some sections may be redundant or inefficient.	Program is fully sensor-driven and responsive; advanced techniques (e.g., variables, counters, or "While" loops) are used to create clean, concise logic; code perfectly matches the information provided in the Engineer's Guide.	
Code Documentation (X1)	Code contains little to no comments or explanation; variable names are generic (e.g., "Motor1", "SensorA") making the logic difficult to decipher for a third party.	Code includes basic comments explaining major sections; variable names are somewhat descriptive; general flow of the program can be understood.	Code is professionally documented; complex logic is explained with clear comments; variables use descriptive names (e.g., "Red_Reject_Arm") that make the code self-explanatory and easy to debug.	
			PROGRAMMING STRUCTURE SUBTOTAL (20 points)	

DEMONSTRATION/INTERVIEW (10 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Technical Articulation (X1)	Team struggles to explain how their system works; responses are vague or "I don't know"; technical terminology is incorrect or missing.	Team articulates their design process clearly; most members can answer questions about the mechanism or code; leadership and collaboration are evident.	Team demonstrates mastery of their system; they effectively use industry terminology to explain specific design and coding choices; every member contributes to the technical discussion with confidence.	
			DEMONSTRATION/INTERVIEW SUBTOTAL (10 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL** (110 points)	

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

AUTOMATED MANUFACTURING SYSTEMS EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator
B. Judges, two (2) or more
C. Assistants, two (2)

MATERIALS

A. Coordinator's packet, containing:
    1. Event guidelines, one (1) copy for the coordinator and for each judge
    2. TSA Event Coordinator Report
B. Stopwatch, one (1)
C. Written problem, one (1) copy per team and judge
D. Engineer's Guide, one (1) copy per team
E. Large room to accommodate teams
F. One (1) table and three (3) chairs per team
G. Interview questions

RESPONSIBILITIES

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.
B. Report to the CRC room and check the contents of the coordinator's packet.
C. Review the event guidelines and check to see that enough personnel have been scheduled.
D. Inspect the area(s) in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
E. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to distribute materials and to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

A. Begin the event at the scheduled time by closing the doors.
B. All participants and judges should be in the room at this time.
C. Late entries are considered on a case-by-case basis and only when the delay is caused by events beyond participant control.
D. In order to compete, participants must be on the entry list or have approval of the CRC.
E. Secure participants' equipment in the area designated.

PRELIMINARY ROUND

A. At the orientation meeting obtain the team/chapter identification numbers from each team captain.
    1. Judges must be present at the orientation meeting.
    2. Review the time limits, procedure, and regulations with teams.
B. Distribute the problem and Engineer's Guide to teams at the beginning of the event.
C. Teams have thirty (30) minutes to plan for their solution.
D. Each team is given three (3) hours to complete the remaining portions of the event.
E. Teams must demonstrate that their device/model prior to leaving.
    1. Judges must observe this portion and shall perform a short interview.
    2. Judges may take notes, but evaluation occurs only after all teams have left the event room.
F. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:
    1. To deduct 20% of the total possible points or
    2. To disqualify the entry

    The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
G. If necessary, manage security and the removal of materials from the event area.